

Secondary Index (Non-Clustering Index)

1. Overview

A **Secondary Index** is an index built on a column that **does not define the physical storage order** of the table.

Since the data file is not sorted on that attribute, this index is separate from the actual data storage.

None

Table storage order $\neq$ indexed column order

This is why it is also called a **Non-Clustering Index**.

2. Simple Meaning

Suppose rows are stored by **id**, but you search by **email**.

None

Stored by id

Search by email

Need a secondary index on email.

3. Why Needed

Without secondary index:

```
SQL
SELECT * FROM users
WHERE email='abc@mail.com';
```

May scan whole table.

With secondary index:

```
None
email → row pointer
```

Fast lookup.

4. Example Table

Users table physically stored by **id**:

id	name	email
1	A	a@mail.com
2	B	b@mail.com

3	C	c@mail.com
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Secondary index on email:

None

a@mail.com → row1

b@mail.com → row2

c@mail.com → row3

5. Why Called Secondary

Historically:

- One main ordering index already exists
- Additional indexes are “secondary”

Modern systems can have many indexes.

6. Can Be on Key or Non-Key Attributes

A. Key Attribute

None

passport_no

email

username

(unique values)

B. Non-Key Attribute

None

department

city

status

(many rows may share same value)

7. Dense Index Nature

Secondary indexes are commonly **dense**, meaning:

None

Every search-key value has index entries

Often one per row, or one key pointing to row-list.

Because rows are not sequentially grouped physically.

8. Example Non-Key Secondary Index

Employees table:

id	name	dept
----	------	------

1	A	HR
2	B	IT
3	C	HR

Secondary index on dept:

None

HR → row1,row3

IT → row2

9. Secondary vs Primary / Clustered Index

Feature	Primary / Clustered	Secondary / Non-Clustering
Defines storage order	Yes	No
Physical row arrangement	Same as index	Separate
Number usually	One	Many
Good for extra search paths	Moderate	Excellent

10. Real DB Mapping

Modern terminology:

Classical Term	Modern Equivalent
Secondary Index	Non-clustered index

Used in:

- PostgreSQL
- MySQL
- Microsoft SQL Server

11. Internal Structure

Often implemented using:

- B+ Tree
- Hash Index
- Bitmap Index (analytics)

12. Benefits

Fast WHERE lookups

Faster JOINS

Multiple search paths

Better query latency

13. Costs / Trade-offs

Extra storage

Slower INSERT/UPDATE/DELETE

More index maintenance

Too many indexes hurt writes

14. Example Query Gains

SQL

```
SELECT * FROM users WHERE email='x@mail.com';
```

```
SELECT * FROM orders WHERE status='PAID';
```

```
SELECT * FROM employees WHERE dept='IT';
```

15. HLD Perspective

In scalable systems, secondary indexes are important for:

None

Users by email

Orders by user_id

Products by category

Events by timestamp

16. Interview Answer Example

Q: What is a secondary index?

A secondary index is a separate index on a column that does not determine physical row order, enabling fast searches on alternate attributes like email, city, or status.

17. Short Revision Notes

None

Secondary Index:

- Non-clustering index
- Separate from row storage order
- Used for extra search columns

Usually dense

Good for WHERE queries

18. One-Line Summary

A secondary index is a non-clustered searchable structure built on columns other than the table's physical ordering key, providing fast alternate lookup paths.